

It is very important to LEARN the table on p. 403 of Stewart.

Also: Do examples 1 to 5 on p. 404-405.

Some mixed examples: Do these on your own first.

$$1) \int (\sqrt{x} + \frac{1}{\sqrt{x}}) dx$$

$$2) \int x \sqrt{1+x^2} dx$$

$$3) \int \sin x \cos x dx$$

$$4) \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

$$5) \int \cos^2 4x \sin 4x dx$$

$$6) \int e^{\tan 8x} (\sec^2 8x) dx$$

$$7) \int \cos x e^{3 \sin x} dx$$

$$8) \int \frac{\cos x}{1+2 \sin x} dx$$

$$9) \int \frac{\sec x \tan x + \cos x}{\sec x + \sin x} dx$$

$$10) \int \frac{x}{1+x^2} dx$$

$$11) \int \frac{1}{1+x^2} dx$$

$$12) \int \frac{x}{(1+x^2)^2} dx$$

$$13) \int \frac{x^2}{1+x^6} dx$$

$$14) \int \frac{1}{\sqrt{9-x^2}} dx$$

$$15) \int \frac{1}{4+x^2} dx$$

$$16) \int \frac{x^2}{x^2+1} dx$$

$$17) \int \frac{x^4+1}{x^3} dx$$

$$18) \int x^2(1+x^2)^2 dx$$

$$19) \int \operatorname{cosec} x \cot x e^{\operatorname{cosec} x} dx$$

$$20) \int \frac{\sinh(\ln x)}{x} dx.$$

Solutions:

$$1) \int \sqrt{x} + \frac{1}{\sqrt{x}} dx = \int x^{\frac{1}{2}} + x^{-\frac{1}{2}} dx = \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C.$$

$$2) \int x \sqrt{1+x^2} dx = \int x(1+x^2)^{\frac{1}{2}} dx = \frac{1}{3} (1+x^2)^{\frac{3}{2}} + C$$

ALWAYS TEST! $\frac{d}{dx} \left(\frac{1}{3} (1+x^2)^{\frac{3}{2}} + C \right)$

$$= \frac{1}{3} \times \frac{3}{2} (1+x^2)^{\frac{1}{2}} \times 2x + 0$$

$$= x \sqrt{1+x^2}$$

$$3) \int \sin x \cos x dx \quad (3 \text{ ways to do this})$$

$$= \frac{1}{2} \sin^2 x + C$$

Test: $\frac{d}{dx} \left(\frac{1}{2} \sin^2 x + C \right) = \sin x \cos x.$

$$4) \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = -2 \cos \sqrt{x} + C \quad \frac{d}{dx} (-2 \cos \sqrt{x})$$

$$= -2 (-\sin \sqrt{x}) \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{\sin \sqrt{x}}{\sqrt{x}}$$

$$5) \int \cos^2 4x \sin 4x dx$$

$$= -\frac{1}{3} \times \frac{1}{4} \cos^3(4x) + C$$

$$= -\frac{1}{12} \cos^3(4x) + C.$$

$$\frac{d}{dx} \left(-\frac{1}{12} \cos^3(4x) + C \right)$$

$$= -\frac{1}{12} (3 \cos^2(4x) \times (-\sin 4x) \times 4)$$

$$= +\cos^2(4x) \sin(4x)$$

$$6) \int e^{\tan x} (\sec^2 x) dx$$

$$= \frac{1}{3} e^{\tan x} + C.$$

$$7) \int \cos x e^{\sin x} dx$$

$$= \frac{1}{3} e^{\sin x} + C$$

Test!

$$8) \int \frac{\cos x}{1+2 \sin x} dx = \frac{1}{2} \ln(1+2 \sin x) + C.$$

$$9) \int \frac{\sec x \tan x + \cos x}{\sec x + \sin x} dx = \ln |\sec(x) + \sin(x)| + C$$

$$10) \int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2) + C$$

$$11) \int \frac{1}{1+x^2} dx = \arctan(x) + C$$

$$(12) \int \frac{x}{(1+x^2)^2} dx = \int x (1+x^2)^{-2} dx$$

$$= -\frac{1}{2} (1+x^2)^{-1} + C \quad \text{Test!}$$

$$(13) \int \frac{x^2}{1+x^6} dx$$

$$= \int \frac{x^2}{1+(x^3)^2} dx$$

$$= \frac{1}{3} \arctan(x^3) + C \quad \text{Test!}$$

$$\underline{\underline{NB}} (14) \int \frac{1}{\sqrt{9-x^2}} dx$$

$$= \int \frac{1}{\sqrt{9(1-x^2/9)}} dx$$

$$= \int \frac{\frac{1}{3}}{\sqrt{1-(\frac{x}{3})^2}} dx = \arcsin\left(\frac{x}{3}\right) + C.$$

$$\underline{\underline{NB}} (15) \int \frac{1}{4+x^2} dx$$

$$= \int \frac{1}{4(1+\frac{x^2}{4})} dx$$

$$= \frac{1}{2} \int \frac{\frac{1}{2}}{1+(\frac{x}{2})^2} dx = \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C.$$

$$\underline{\underline{NB}} (16) \int \frac{x^2}{x^2+1} dx$$

$$= \int \frac{x^2+1-1}{x^2+1} dx = \int \frac{x^2+1}{x^2+1} - \frac{1}{x^2+1} dx$$

$$= \int 1 - \frac{1}{x^2+1} dx$$

$$= x - \arctan(x) + C.$$

$$(17) \int \frac{x^4+1}{x^3} dx$$

$$= \int \frac{x^4}{x^3} + \frac{1}{x^3} dx = \int (x + \frac{1}{x^3}) dx = \dots$$

$$(18) \int x^2(1+x^2)^2 dx = \int x^2(1+2x^2+x^4) dx$$

$$= \int (x^2+2x^4+x^6) dx$$

$$= \frac{1}{3}x^3 + \frac{2}{5}x^5 + \frac{1}{7}x^7 + C.$$

$$(19) \int \operatorname{cosec}(x) \cot(x) \delta^{\operatorname{cosec}(x)} dx$$

$$= \frac{-\delta^{\operatorname{cosec}(x)}}{\ln \delta} + C$$

$$(20) \int \frac{\sinh(\ln x)}{x} dx$$
$$= \cosh(\ln x) + C$$

Test:

$$\frac{d}{dx}(\cosh(\ln x) + C)$$
$$= \sinh(\ln x) \times \frac{1}{x} + 0$$
$$= \frac{\sinh(\ln x)}{x}$$

ALWAYS, ALWAYS, ALWAYS test your answer!!